

Advanced contribution of IoT in agricultural production for the development of smart livestock environments

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ABSTRACT

Many areas of contemporary life need the use of Artificial Intelligence (AI), Machine Learning (ML), and the Internet of Things (IoT) for analysis and comprehension. The application of AI and ML in livestock farming has led to an improved understanding of animal behavior and discomfort, the prevention and management of diseases, and the efficiency of the farmer's economic choices. Digital twin technology is a particularly promising field that builds on AI and is already being utilized to increase efficiency and decrease costs in cattle livestock agriculture. A Digital Twin (DT) is a constant-update numerical copy of smart livestock environments. The proposed research is IoT, and Digital Twins in Livestock Farming (DTLF) might expand extensive Precision Livestock Farming (PLF), technology and tools use, and farm animal health and well-being. Global Positioning Satellite (GPS), Radio Frequency Identification (RFID), AI, and even ML are only some current technologies that may be used for automated tracking of individual animal whereabouts. There is promise in using these methods and innovations to monitor and evaluate animal welfare, but these techniques are difficult. Just like other revolutionary innovations, the utilization of the DT mechanism will improve disease analysis prediction for individual farms by a factor of 92%. The responsiveness of DTLF will be a 92% response rate in analyzing the Heartbeat, a 94% effectiveness rate in analyzing the Temperature Range, and a 94% overall effectiveness in evaluating the Humidity Range.

Introduction

Agriculture serves as the backbone of every country that is both economically and socially advanced. Nowadays, most people agree that forestry, fruit production, dairy, poultry, beekeeping, etc., belong to agriculture [1]. In response to market pressure and demand for milk products, the size of dairy farming has shifted over the last several years, with a tendency toward bigger, more intense, profit-driven companies. The demand for the expansion of dairy farming has resulted in a rise in interest in the technologies that make agricultural automation and other comparable feats possible [2]. The cattle business is essential to global trade for meat, leather, and dairy. The function of domesticated animals, like cows, buffalos, sheep, goats, etc., is significant in rural communities for nutrient cycling and food production on modern, sustainable farms. Farmers are taking a major financial hit due to the widespread sickness affecting their livestock. Studies on a health monitoring system to track the spread of sickness and the behavior of individual animals are planned to help avoid similar disasters [3]. There is significant investment and development in the field of high-tech cow

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production. In the past, many people around the globe kept cows on their farms and ranches. In small-scale farming, the owner's familiarity with the animal's health and other statuses makes such monitoring very easy. Manual supervision techniques are quite challenging to implement for intensive livestock. Cattle management may benefit from behavior activity monitoring in several ways [4]. The challenges in employing these strategies and tools to assess animal well-being can be broken down into two categories: (a) the technical difficulties inherent in applying the technologies, and (b) the ethical implications of employing them. The accuracy and dependability of the technologies presents the greatest technical challenge. Imaging technologies, for instance, need to be able to distinguish between conditions that are indicative of animal welfare and those that are not, and automated monitoring devices need to be able to accurately identify and measure animal behaviors. Data collection methods need to be stable and consistent over time. These technologies' widespread adoption has prompted ethical concerns about invasion of privacy and their potential for abuse. For instance, decisions made with the help of the information gathered by these technologies might not always be in the animals' best interests. It is really possible that the collected information could be used to target and harm certain kinds of animals. Furthermore, there are concerns that animals may be used as data sources due to the prevalence of such technologies.

Furthermore, the difficulties encountered are relevant to dairy farmers everywhere: a need to improve resource efficiency, and green their operations. They adopt trimming technology developments that may give consumers in-depth information while protecting product quality. The deployment of cheap technology for monitoring and optimizing water and energy resources might benefit large-sized farms in developed and underdeveloped countries [5].

With the advent of IoT, GPS systems, and another wearable tech, remote monitoring, wireless sensor networks, and control mechanisms have all been transformed to become more concise and accessible. In order to update health information or to manage the device using a smartphone, all of these approaches need internet access and interfaces. The range of sensors and gadgets that can be integrated into farming systems has expanded greatly. Overall, technological progress has aided the expansion of IoT, leading to the birth of several novel concepts like PLF, which tracks and responds to the variability of farming information received by sensing devices [6]. Digital farming employs data technology to perform a more in-depth analysis of the agricultural system by factoring in position, previous, concurrent, and predictive data [7].

The IoT paradigm [8] is crucial in monitoring resources in Smart Farming by linking various structures, equipment and vehicles, and even living beings [9]. In order to monitor farms, conduct surveys, and provide data to farmers for sensible farm management plans, the IoT employs drones, robots, sensors, and computer imagery in conjunction with ever-improving ML algorithms [10]. When it comes to physical assets, digital twins may reflect changes in near-real time, with lag times of only milliseconds to minutes at most. As a bonus, digital twins can gather and analyze more data than traditional computer simulations. Because of this, digital twins can create more accurate scenarios. In a number of ways, digital twin technology can make cattle livestock farming more efficient and less expensive. Digital twins have several advantages for the livestock industry. First, they allow farmers to keep tabs on their animals in real time, analysing their health and providing them with data-driven insights that can speed up the decision-making process. For instance, DTs can be used to identify the earliest stages of a disease, giving farmers a chance to intervene before the problem escalates. Data collected by DTs can be used to fine-tune other aspects of livestock management, such as the amount of food and water each animal receives, ultimately leading to happier, more productive animals. Last and not least, DTs can help find and eliminate sources of waste and inefficiency in the manufacturing process, saving money.

However, input data, such as detector and video information [11], is needed for an instantaneous tracking system to identify the sickness. The IoT, AI, ML, blockchain, and robots are just some technologies that may provide an advanced, secure digital solution in real-time. Cattle connected with IoT technology sensors collect sensory data sent to a vision-based device. Applying a trained AI or ML technique to this data will allow illness detection based on input qualities and disease detection results to be sent to the visual device [12].

RFID, GPS systems, and sensors have a wide variety of uses. RFIDs have several applications, including physical distribution, access restrictions, animal identification, etc. Among the various uses for GPS technology is in-vehicle navigation structures. Sensor networks, in particular, are seeing a tremendous expansion in their range of potential uses. Some potential use cases are healthcare, the military, and domestic settings. Sensors, for instance, have several applications in military contexts, including surveillance, observation, and targeting. Sensor nodes are not limited to catastrophe monitoring and stock management [13]. Ecosystems of animals and plants are studied with the use of sensor networks. In addition, a cow-tracking service is being created using GPS and a satellite system. The use of RFID, GPS, and sensors to monitor animal and staff movement in shelters has not yet been thoroughly researched or offered as a service [14]. WIFI, Bluetooth, cellular technology, and ZigBee are some wireless options that may be used to convey data about the animals on a farm. IoT-based livestock applications, such as those for monitoring animal health and tracking their movements, rely on the interconnectedness of many storages and servers that make up cloud architecture [15]. The majority of animal tracking today is done via global positioning system. The precise location of an animal can be determined and followed in real time. Technology like this relies on tethering a GPS-enabled device to the animal, which then communicates with satellites to pinpoint its precise location. Similar to GPS, RFID can be used to follow the movements of animals. It makes use of tags that can be read by a reader and are attached to the animal. Although it is not as precise as GPS, this technology can still reveal valuable insights into an animal's whereabouts, how it moves, and how it behaves. Artificial intelligence can be used to decipher the information gathered by GPS and RFID collars and other animal tracking devices to shed light on the animal's habits location information. This information can help find patterns in the animal's behavior and pick out potential dangers. Models can be developed using ML that accurately foretells how an animal will act in a given situation. This can help us get a handle on the animal's habits and habits, as well as give us some insight into potential dangers.

In general, the technology employed for the automated tracking of individual animals is determined by the requirements of the task at hand.

Thus, this study presents an intelligent animal situation monitoring service for zoos and aquariums using RFID, GPS, and sensors.

Sensor nodes would be used to measure animal body temperatures, GPS would be used to track animals, and RFID would be used to identify animals. A prototype of the proposed tracking service is then developed and tested when a service scenario is presented. The suggested system may update a website in real-time with data like animal body temperature, position, etc. Numerous sensors, either worn by the animals themselves or strategically placed in their environments, can record data such as their body temperature and location in real time. These sensors can detect the animals' temperatures and localize them in space. It is possible for the sensors' data to be transmitted wirelessly to the website in real time. When this information is submitted, the website can make instantaneous adjustments. Even if animals can get out of their enclosures, technology can locate them. The main objectives of the proposed paper are followed as.

- The proposed study is an IoT-integrated Digital Twins in Livestock Farming (DTLF) that can greatly increase the scope of Precision Livestock Farming (PLF), the application of technology to monitor the welfare of farm animals.
- DT and some sensors, such as GPS, and RFID, are used for automated tracking of individual animal locations and welfare.
- The suggested study is expected to provide findings about efficiency, heart rate, disease detection rate, humidity rate, and cattle temperature.

The suggested research is briefly summarized below.

Section 1 covers the literature review; Section 2 provides a thorough description of IoT-DTLF; Section 3 presents the results; Section 4 offers a summary and last observations.

Literature survey

Alonso et al. [16] examined the dairy sector and the way it may better respond to the demands of the modern market by improving its resource efficiency, environmental friendliness, transparency, and safety measures. With the advent of modern technologies like the Internet of Things (IoT), Edge Computing (EC), and Distributed Ledger Technologies (DLT), the entire value chain can be digitized, providing consumers with previously unavailable levels of information about the final product while also guaranteeing its quality and safety. IoT and DLT provide source observing and traceability throughout the significance sequence, allowing producers to improve operations, give the source of the food, and assurance of the value to customers in smart farming settings. As opposed to storing all of the big data produced by IoT procedures in one centralized location, EC handles such information at the system's edge and paves the way for faster reaction times, better QoS, and more secure services. With the help of cutting-edge Global Edge Computing Architecture (GECA), the author of this work provides a platform designed for instantaneous surveillance of milk production and feeding cereal, as well as tracking and efficiency of the different production processes. Real-world deployment and testing on a dairy farm prove the platform's efficacy in lowering data loss due to the introduction of EC.

Symeonaki et al. [17] introduced distributed IoT middleware as the foundation of a responsive and adaptive service-oriented Integrated Farm Management (IFM) system designed to allow smart livestock farming in light of the industry's special needs. This middleware will help with the administration, processing, and sharing the massive amounts of heterogeneous data points created in livestock facilities. The suggested IoT middleware has a versatile ontology-based framework for modeling and reasoning, making it possible to use a context-aware approach. A case study was conducted to evaluate the IoT middleware in an actual setting to implement smart thermal environment management in a moderately large pig farm. The assessment findings suggest that the system performs well in terms of computing performance and ontological coherence, consistency, and efficiency.

With a rising number of connected IoT components, present frameworks' scaling capabilities and dispensation control would be attained. For the issue, Deniva et al. [18] proposed a particularly built structural design based on Amazon Web Services (AWS) for monitoring cattle utilizing Cyber-Physical Systems (CPS) and IoT apparatus. As a result, it's important to discover a way to keep up with the increasing needs. The AWS cloud services being used are detailed in-depth, and their specific functions are explained about the application being used. The created architecture for data processing has been put through a stress test to demonstrate its reliability. When used in conjunction with Amazon's services, the existing architecture has been experimentally shown to be capable of processing the required data volume and allowing for the implementation of automated scaling mechanisms inside the CPS/IoT infrastructure.

Pan et al. [19] developed the research center on the issue of the present industrial IoT system for cattle farming being too inflexible, difficult to extend, and having poor network performance. In this research, the researcher offers a framework for a unified IoT system for intelligent cattle farming that uses RESTful (REST: Representational State Transfer) web services. First, the current state of smart cattle farming IoT system research and development worldwide is presented. Then, two popular Web services technologies, Simple Object Access Protocol (SOAP)-based WS and REST, are contrasted, and the circumstances in which they shine are outlined. Finally, a prototype IoT system for intelligent animal husbandry is shown, emphasizing chicken coop environment monitoring and control using Restful Web Services. The prototype's testing results demonstrate that using Restful Web Technology may boost application development efficiency and offer a workable system development paradigm for a smart cattle farming IoT system. Sensors attached to livestock and a mobile app that displays information about the animals' exact location, health, and behavior could form the basis of a prototype of the proposed tracking service. The prototype could feature a platform for real-time monitoring and analysis of sensor data. This could help farmers learn more about their animals' habits, spot potential issues, and take corrective action.

Shang et al. [20] discussed that the breeding environment for livestock and poultry significantly influences animal development, illness prevention and treatment, and product quality. Collecting data on the breeding environment is key to successful cattle and poultry breeding. A 16-channel wireless monitoring system for use in a cattle or poultry breeding setting was created, complete with schematic designs and printed circuit board (PCB) artwork. Programs were built for the microcontroller unit (MCU) to obtain data

about the breeding environment, the heartbeat mechanism, and the reading of sensory data in a continuous loop. An organizational chart for the apparatuses to monitor the environment was drawn up. Using system integration technologies, we created IoT-based devices for monitoring the open-field breeding environments of livestock and poultry. Eleven farms with cattle and chickens used the sensors in a trial run. The equipment's steadiness and malleability were examined. Incorporating this device's design and development into livestock and poultry farms might give useful technological assistance for data collection, automation, and cutting-edge management.

Germani et al. [21] presented the planning, execution, and results of an IoT hardware and software architecture designed for the round-the-clock observation of animals in stables and on pastures. To accommodate these varying conditions, we've developed a web service architecture that stores, processes, and displays data collected over LoRa LPWANs. Instead of using the Low Power Wide Area Networking (LoRaWAN), Media Access Control (MAC), which does not offer Listen-Before-Talk (LBT), researchers have proposed using a specialized MAC layer with LBT-based carrier-sense various access with collision warning. The designed system runs on commercially available hardware, and its performance has been measured using a C++ event-based simulator. Early findings from our HW implementation in the field have confirmed the stability and dependability of the intended system. Livestock management as it is practiced today can be time consuming and ineffective. Constantly keeping close tabs on an animal's whereabouts and condition can be challenging as well. If animals become ill or die prematurely as a result of this, it can have a negative impact on productivity and revenue. Furthermore, the current method places a premium on the farmer's knowledge, which can result in expensive mistakes.

Building Digital Twins in Livestock Farming (DTLF) livestock may help with these drawbacks. Automated systems called "Digital Twins in Livestock Farming (DTLF)" "track and record data about the animals' well-being, habitat, and diet. Increased productivity and sustainability in agriculture are possible with this information. Each animal's current state of health and productivity can be monitored in real time with the help of such systems. Optimization of food intake, early disease detection and prevention, and behavior monitoring are all possible due to this information. These systems can help farmers save money on labor expenses while increasing output per worker.

a) Introduction to the Architecture of an IoT technology for Livestock:

The architecture of IoT networks proposes a framework for identifying the material components of livestock, observing and elaborating their main operating beliefs with applied methods. Fig. 1 depicts the layered architecture of an IoT for livestock systems.

The Internet of Things (IoT) enabled smart livestock agricultural infrastructure is a system that combines modern technology with time-honored methods of livestock farming. It's meant to help ranchers keep an eye on and manage their herds more cheaply and effectively. Animal behavior, feed and water intake, temperature, and other environmental factors can all be monitored in real time with the help of this system. This aids farmers in making better, more proactive choices that increase animal well-being, output, and income. The system helps to improve animal welfare while decreasing the price of labor-intensive manual labor. Farmers now have a powerful tool in the form of the IoT-enabled smart livestock agricultural infrastructure, which allows them to better manage their herds while also reducing their livestock's negative effects on the environment.

APPLICATION LEVEL

MQTT, COAP, AMQP, XMPP, SSL

TRANSPORT LEVEL

UDP, TCP

NETWORK LEVEL

IPV4, IPV6, RPL

PHYSICAL LEVEL

ZWave, EPC-Global, LTE

Fig. 1. IoT-enabled smart livestock agricultural infrastructure.

• APPLICATION LEVEL:

Application layer techniques may be scaled up or down depending on the needs of the network. These protocols include Message queuing telemetry support (MQTT), Constrained Application Protocol (CoAP), Extensible Messaging and Presence Protocol (XMPP), Advanced Message Queuing Protocol (AMQP), and Secure Sockets Layer (SSL). The purpose of this layer is to facilitate the transfer of data between the various levels of the design. Information about a cattle's health is collected by android websites or apps and sent to the application level. MQTT is a bidirectional, low-bandwidth connection that provides reliable health monitoring of livestock. Transport Protocol Constrained Application Protocol (CoAP) was developed with the limitations of IoT network architecture in mind. While Secure Sockets Layer (SSL) protocols are used for safety in cattle settings, AMQP maintains a subscribe design.

• TRANSPORT LEVEL:

The Transport layer acts as a bridge between the network and application levels to ensure smooth data movement between devices. The major function of this layer is to gather and examine all sensor data about cattle. The Transmission Control Protocol (TCP) ensures the integrity of the data collected. User Datagram Protocol (UDP) offers faster transmission speeds at the expense of reliability. In contrast, the TCP is notoriously sluggish. Thus, the UDP and TCP protocols are used in various contexts depending on the needs of the given application.

• NETWORK LEVEL:

The Network Layer is the principal layer responsible for transmitting data gathered from livestock applications (such as those concerned with monitoring animal health, tracing animals, keeping tabs on pets, etc.) to the appropriate application layer. The IoT network architecture concept uses the IEEE 802.15.4 protocol for data transport between livestock sensors and hangable devices utilizing 6LoWPAN and IPv6 devices. As a bonus, UDP is used for two-way communication between sensing devices and the cloud. RPL routing protocols support multiple traffic flows, and routing metrics like connection quality, greater computational cost exchange, and battery state are considered.

• PHYSICAL LEVEL

The physical level is the lowest in the IoT system. It is in charge of acting on and sensing the animal's health-related indicators like temperature, rumination, cow movement, etc. Z-wave and LTE are the standards present at this level. The IEEE 802.15.4 standard is the most extensively adopted and utilized of the four mentioned here because of its simplicity, low power requirements, and cheap cost. Accordingly, several protocols, including ZigBee, Wireless HART, and ISA100, have accepted this standard. In addition to IEEE 802.15.4, other technologies were utilized to transmit data directly via Internet Protocol (IP).

a) Livestock farming with the use of digital twins:

Livestock farms may benefit greatly from receiving specific, quick, accurate intelligence from their physical assets. Historically, farmers have relied on human observations, experience, and projections of the weather and market prices (dynamics). In the future, farming may instead be based on instantaneous information controlled by AI analysis to make good business choices, increase cattle welfare, and enhance the return from natural crops. One further perk is that they need not be physically present at the farm to use this feature. The physical property of a livestock farm is integrally connected to its routine tasks and operations. Assets can be outfitted with sensors and other technologies to track data like temperature, feed consumption, water use, and animal health. The farm can use this information to conduct an in-depth analysis of its performance and make informed, productive choices moving forward.

Assessing the condition of the farm's infrastructure can indeed shed light on the livestock population's general well-being. The ability to detect environmental changes and animal health abnormalities is included. Improved farm management and higher yields are two outcomes that may result from considering these factors.

Last but not least, a livestock farm's management can be improved with the help of reliable data gathered from physical assets. The automation of routine tasks on a farm, such as maintaining feeding schedules and keeping track of animal health, can increase efficiency and save money. As a result, the farm may be able to increase its profit margins.

b) PLF: A Step towards Digital Twins.

While DT technology in livestock farming remains developing, PLF has employed current methods to enhance cattle health management and production output. PLF is becoming more popular because of rising labor costs and expectations of cattle business automation. The innovative approaches to animal agriculture problem-solving that have been proven via PLF clear the way for the widespread use of DT in cattle farming. PLF has previously used animal sensors that may be worn on the body, a crucial instrument for advancing digital twin technology. Below are some concrete applications of sensor technology that might serve as digital twins in animal production. Fig. 2 demonstrates the elaborate structural design of IoT-based DTLF technology.

» Sensors that detect infrared radiation and use it to calculate temperatures are called thermal infrared sensors.

- » A belt is worn around the chest (much like a halter) that measures the animal's thoracic and abdominal motions to determine its respiratory rate.
- » Immuno sensors can analyze bodily fluids from animals to determine the levels of hormones like cortisol and lactate. It also leads to non-invasive diagnostic procedures.
- » Photo Plethysmo Graphy (PPG) is a technique that uses infrared light to assess microcirculation and, in turn, identify changes in plasma capacity in the tissue. It's a cheap and painless technique to monitor blood volume.

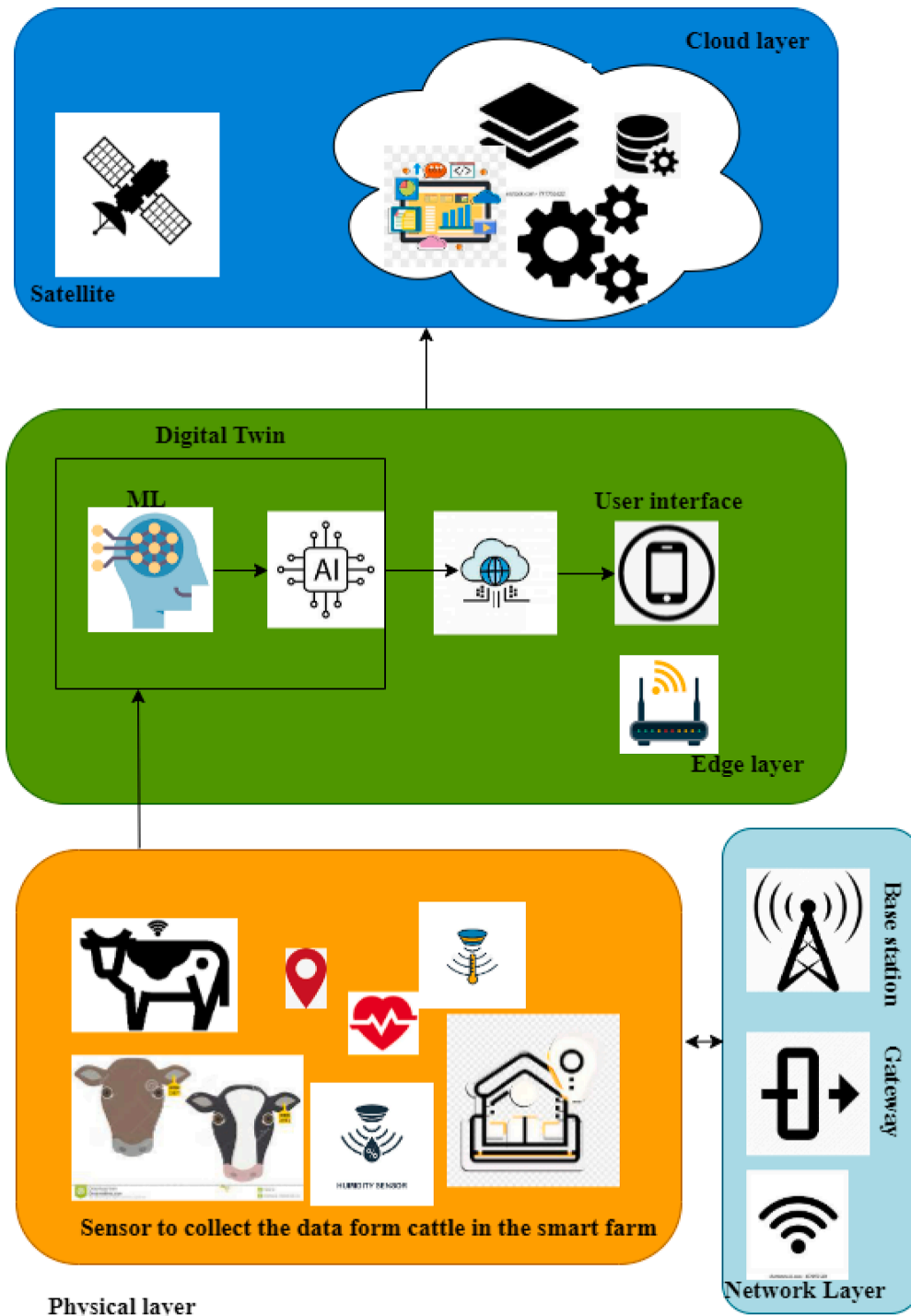


Fig. 2. demonstrates the layered structural design of IoT-based DTLF.

- » Farmers may better manage stressed animals with a noseband sensor like the Rumi Watch, which tracks feeding and rumination patterns in dairy cows.
- » Monitor the watering habits of large groups of animals; water flow sensors may be installed. Their advice is often believed to be reliable.
- » Accelerometers can detect the acceleration forces exerted by a moving animal using electromechanical signals. When it comes to tracking animal behavior, they are quite reliable.
- » Each animal's daily step count may be measured objectively by a pedometer, and the resulting distance can be calculated with the help of an algorithm. They are useful in detecting lameness and stress.
- » Intraluminal bolus sensors that are wireless and may be introduced via the esophagus must keep on technologically advanced to observe rumen and reticulum temperatures and pH levels. Ruminant acidosis and hypocalcemia are two disorders that may be detected with their aid.
- » Lastly, these sensors, along with others, may be used in tandem to detect a variety of stressors, diseases, and pains in the body.

• Observing the Livestock's mobility While Grazing:

GPS has yielded previously unavailable insights. Animal migrations have been mapped out with their help. Researchers in the livestock industry have utilized GPS data from free-grazing tests to better understand the behavioral distinctions between the animals. The problem of maintaining experimental unit autonomy arises when attention is shifted from the surrounding environment to the animal itself. The practice of social facilitation threatens the person's autonomy inside a group. An individual's actions are facilitated by their social surroundings, a phenomenon known as social facilitation. Depending on the context, it can either strengthen or weaken a person's independence. A person's independence may decline if they are subjected to persistent social pressure to act in ways that are deemed acceptable by the majority. On the other hand, a person's independence can be bolstered through social facilitation if they gain self-assurance and start speaking up for themselves. In behavioral investigations, using spatial modeling techniques to interpret GPS data allows researchers to ascertain the degree to which data from a single animal may be considered independent. Researchers could examine animals' reactions to organizational processes or ecological disturbances by employing position and mobility data gained from GPS data. GPS has helped them quantify the ecological impact of feeding animals. Landscape rejection and choosing behaviors may be more easily identified when satellite-derived remote-sensing data is combined with Navigation terrain choice indexes. Some of the behaviors are 1. Image segmentation, Feature extraction, Image compression, Classification, Change detection, Pattern recognition, Radiometric correction, Geometric correction, Edge detection, Pixel-level analysis.

The collar belting assembly and electronics container try to compensate for the Clark ATS+ collar rigging. The collar's electronics are stored and protected within a polycarbonate, watertight, two-part box called an enclosure. The device's container may be attached around like an animal's neck with the help of the belting arrangement, which also helps to shelter and secure the GPS and satellite broadband antennae and related connections. A GPS collar's 8-channel receiver can keep tabs on eight satellites simultaneously. Data is transferred to onboard non-volatile memory with random access to track up to 5028 unique values for adjustments. Each log entry consists of a cow identifier, day, hour, coordinates (location information), altitude, an accuracy dispersion level, whether the fix is in two dimensions, the current atmospheric temperature, and satellite data used for divergence adjustment. Collars may be used to keep track of animals and their movements due to their built-in GPS and data storage capabilities.

Although Radio Frequency Identification (RFID) was created for person detection, it has now been useful in livestock tracking and location. The two main components of RFID systems are the tag or transponder and the reader or transceiver used to read and write information from the transponder. It's possible to use active, passive, or hybrid RFID tags. The reading range for passive and semi-passive RFID is between 10 cm and 3 m since the data is sent by reflection or modification of the electromagnetic field created by the reader. Semi-passive RFIDs only draw power for their sensors and processing circuits from their batteries. Since it contains a battery, an active RFID can send and receive signals up to 100 m away with greater clarity, but it also needs to include more electrical parts, making it quite bulkier. The detection efficiency of the device is heavily impacted by factors such as the tag's and reader's ideal location, the animal's speed and the proximity of body tissue to the transceivers. RFID-Radar goes beyond just recording a reader's existence to compute the position of transponders that interface with it.

For this reason, it may be used as an RFID-based locating system for indoors. When placed near a feed bunk, water source, or mineral lick block, RFID devices may identify when an animal is eating, drinking, or taking a supplement. Several cattle spent a disproportionate amount of time just standing near the brush. Thus the RFID ear tag recordings of brush closeness were not yet a credible measure of grooming behavior.

• Learning About the Development of Dairy Cattle:

DT may be made and researched to understand more about the development of a dairy cow from a calf to an adult. Multi-agent innovation systems integrate sensor and longitudinal data to create morphological features in livestock. The multi-agent planning process for animals and the management of life-cycle phases and production cycles may benefit from data collected on the animal's structure, behavior, and biological factors.

• AI-Powered Livestock Surveillance through Computer Vision:

■ **Computer vision:** The study of making computers perform visual tasks on a level with humans, such as object recognition and analysis, by simulating at least part of the complexities of the human visual system. The computer vision issue seems easy since even

infants can easily solve it. AI has derived an extensive manner in current centuries. It can now compete with and surpass humans in a wide range of activities that require recognizing and classifying objects. Recent developments in deep learning, neural networks, and A.I. are to blame for this. The vast quantities of data we now create are used to educate computers, helping to accelerate the field of computer vision. OpenCV is an extensive open-source technique for ML and picture analysis. It now plays a crucial role in modern systems' ability to function instantaneously. It may be used to find people, items, and even handwriting in videos and photos. OpenCV's array structure may be analyzed using Python and other tools like Numpy. Statistical methods and vector space are used to identify a wide spectrum of visual features.

■ Artificial Neural Network (ANN):

The field of artificial intelligence encompasses deep learning technologies like ANN. Deep Learning (DL) is an ML subfield that uses various neural network architectures. Since these algorithms are modeled after human cognitive processes, many scientists see them as our greatest hope for creating genuine AI. A sequence-to-sequence model is composed of two recurrent neural networks. The input and the output are processed by a device known as an encoder and a decoder. This approach shines when there is a discrepancy between the input and output data lengths. The most common uses of sequence-to-sequence models are in machine translation, chatbots, and question-answering systems. Let's have a look at the foundation of any Neural Network in Fig. 3:

- **Input layer:** The layer at which users feed information into the model is called the input layer. The input has exactly as many characteristics as neurons in the input layer.
- **Hidden layer:** The input features are passed on to the hidden layer(s), where further processing and actions are carried out. There may be many concealed levels. Layers perform arithmetic operations such as matrix multiplication, convolutions, pooling, etc., and an activation function is also applied.
- **Output layer:** This is where the input from our model is transformed into probability scores using a sigmoid or softmax function, which is then output.
- **Input image layer:** Using a picture as input, you may build a layer for processing visual data. An image input layer, which applies data standardization, is responsible for feeding images into a network. Use the size input to select a custom picture size. Multiple factors, such as the number of color channels and the dimensions of those channels, determine an image's final size. A picture may have anything from zero to three channels, with a gray scale having one and color images having three.
- **The network's input sequence layer** is where the system accepts data sets. For example, the input data is often longer than the output vector in machine translation. Hence the whole input signal must be supplied first before the destination can be anticipated. Without more contexts, the phrase "sequence to sequence models" often refers to a more involved setup.
- **A layer of Convolution:** It may be thought of as simply applying a filter to an input to produce activation. A feature map may be generated by repeatedly applying the same filter to an input, such as an image, showing where and to what extent a recognized feature is present in the input. Unique among neural networks, convolutional ones may automate the discovery of a significant number of parallel filters that conform to the requirements of a certain predictive modeling issue, such as image classification, using a training dataset as input. Therefore, input images could have unusual qualities that aren't encountered very often.
- **Pooling Layer:** When building a Deep Neural Network (DNN), it is usual practice to incorporate a pooling layer in the middle of two subsequent convolution layers. Decrease the number of network parameters and calculations to improve performance. It progressively decreases the spatial size of the representation to prevent overfitting. Each depth slice is processed independently by the Pooling Layer and then spatially enlarged by the MAX operation. In the most common setup, each depth slice in the input is down-sampled by a factor of two in width and height before being fed into a pooling layer, with 75% of the activations being lost in the process. There can be no more than four inputs for a MAX operation.
- **Classification Layers with the Softmax Function:** For Classification Layers that use Softmax, the process of transforming logits into probabilities involves taking the exponents of each output and norming them by the total of the exponents. Because of this, you may

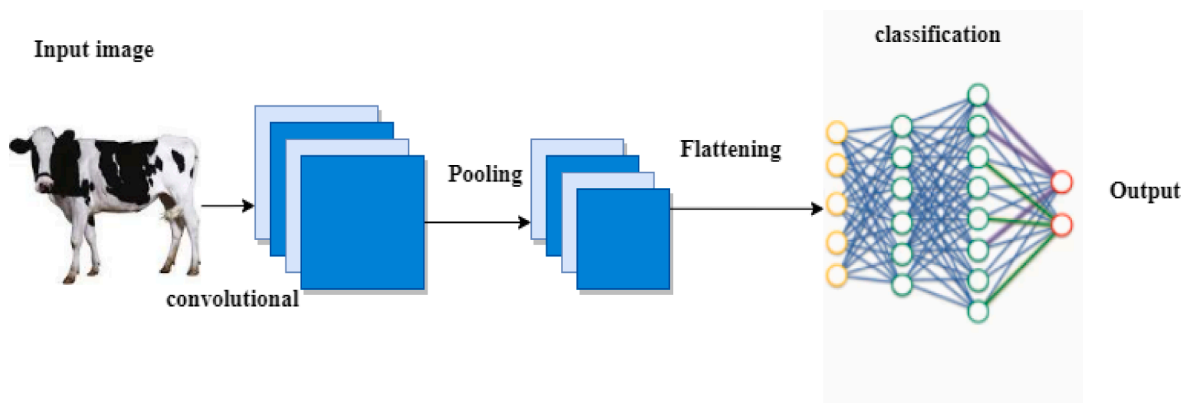


Fig. 3. illustrates the CNN for cattle monitoring.

be certain that the sum of the probabilities in the output vector will equal one. The loss of cross-entropy is typically the result of this kind of issue across several different types. Additionally, CNN-style image classification networks like those used in ImageNet contests utilize a softmax activation function in their final layer.

- **The Application of ML to Train Data:**

ML in SVM's potential to provide competitive or superior results with less training data than competing approaches is one of its biggest appeals. A common use of the SVM method is in the field of automated object identification and classification. The SVM is implemented here in its simplest form as a dualistic classifier, which is used to differentiate between an object and background classes. The method's goal is to efficiently separate an item from its environment using the least available training data. Noise-filled synthetic images are utilized for algorithm evaluation.

For an SVM to function, it must first be trained using data its user provides. It is less dependent on the quantity and amount of training data than ANN. SVM was a binary classifier in its original form, meaning it could only place one of two terms (-1 or 1) on a given test sample. The primary goal of the training procedure is to construct the separation plane according to the characteristics of the training samples. In other words, this hyperplane divides the samples into two groups. For the method to function, the greatest distance between the classes must be maximized by increasing the margin width of the separating hyperplane. Only a subset of the samples is used as the support vector in constructing the hyperplane.

Since the simplest type of SVM, the binary classifier, is used here, the mathematical explanation of the linearly separable situation is provided. The scope of this research does not allow for a discussion of its potential expansion. The training data may be expressed in Eq. (1) for two linearly separable classes.

$$\{i_x, j_x\}, x = 1, 2, \dots, M, j_x \in \{-1, 1\}, i_x \in D' \quad (1)$$

Where i_x stands for the characteristics of the picture and j_x stands for the label of the data. A binary classifier is the most basic implementation of a Support Vector Machine (SVM). This variant of the SVM utilizes a decision boundary to divide observations into one of two classes. Using their proximity to a predetermined line, the data points are classified as either positive or negative. If individuals give the SVM algorithm some data, it will find the hyper plane that most accurately classifies the data. Maximum separation between the two groups yields the best hyper plane. The Eq. (2) for the hyperplane is:

$$I^S + c = 0 \quad (2)$$

Point I on the hyperplane, normal vector X perpendicular to the hyperplane it points in the direction of greatest increase of the function defining the hyper plane, bias c . Let's assume these statements (3, 4) are true,

$$I^S + c \geq 1, j_x = 1 \quad (3)$$

$$I^S + c \leq -1, j_x = -1 \quad (4)$$

Put together and Eq. (4) form the following sentence in (5):

$$j_x(X^S \times I + c) - 1 \geq 0 \quad (5)$$

Then, two hyperplanes, $X^S \times I + c = -1$ and $X^S \times I + c = 1$, may be built with a distance of $\frac{2}{X}$ between them. The next step is to find an Eq. (6) that may help us maximize this margin,

$$\min \left\{ \frac{X^2}{2} \right\} \quad (6)$$

The Lagrangian calculation method is a mathematical strategy for resolving a system of simultaneous equations. Utilizing the Lagrangian function, it finds the best course of action according to the principle of least effort. The Lagrangian method of calculation in Eq. (7),

$$R = \frac{X^2}{2} - \sum_{x=1}^m \beta_x (j_x (X \times I_x + c) - 1) + \sum_{x=1}^m \beta_x \quad (7)$$

Where β is the Lagrange multiplier, the objective is to minimize R here in terms of X and c .

$$\frac{\partial R}{\partial c} = 0 \rightarrow \sum_{x=1}^m \beta_x j_x = 0 \quad (8)$$

$$\frac{\partial R}{\partial X} = 0 \rightarrow X = \sum_{x=1}^m \beta_x j_x I_x \quad (9)$$

The combination of Eqs. (8) and (9) into Eq. (7) yields Eq. (10).

$$R = \sum_{x=1}^m \beta_x - \frac{1}{2} \sum_{x=1}^m \beta_x \beta_y j_x j_y I_x I_y \quad (10)$$

The training process is now optimizing the final equation as a function of $\beta_x \geq 0$. Here, points with $\beta_x \geq 0$ are used as support vectors. In a classification problem using a Support Vector Machine (SVM), the data points closest to the decision boundary are referred to as support vectors. These are the most pivotal pieces of information that will be used to draw the decision boundary and its shape. If there is a binary classification problem, then the support vectors are the data points that characterize the boundary between the two categories. These are the most important pieces of information to consider, even though they can be used to locate and characterize the decision's limits. A classification model's accuracy and robustness can be evaluated with support vectors. It will be recalled that the answers for X and c are in Eqs. (11 & 12),

$$X = \sum_{x=1}^{NV} \beta_x j_x I_x \quad (11)$$

$$c = -\frac{1}{2} X \times (I_n + I_T) \quad (12)$$

Where NV represents the total number of vectors, I_n represents the vectors in the class $j_n = 1$, and I_T represents the vectors in the class $j_T = -1$.

c) Simulated cattle emotions:

Digital twins allow for the real-time monitoring of animal emotions by recreating such feelings in an artificial environment. DT allows their inclusion into future projections of animal behavior and illness management in real-time. This research proposes using this database DT to forecast animal activities by analyzing facial expressions. It will be a method for creating real patterns of videos and photos of continuously shifting facial emotion characteristics of farmed cattle using detector data acquired via bio-feedback systems. It will also be a technique for forecasting and categorizing dairy cattle expressions using Generative Adversarial Network (GAN) classifiers.

d) Early Detection of Serious Livestock lumpy skin Disease:

Diseases are usually only visible when they have progressed to a certain point. Several chances for the illness to spread to other farm animals have passed, raising the probability of an explosion. But digital twins can aid farmers in the early detection of numerous critical illnesses and in isolating infected cattle to prevent further spread.

e) Reducing Animal Feed Waste and Increasing Animal Productivity:

Since so many factors are involved, it is very difficult for farmers and ranchers to determine which feed combinations work best. Still, digital twins may assist in model feed composition changes and execute scenarios to determine the best nourishment consumption

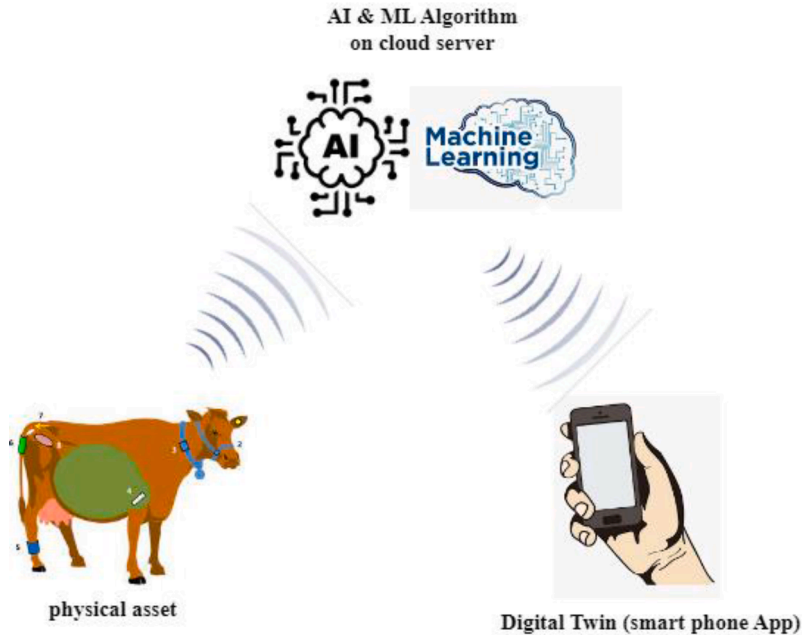


Fig. 4. Demonstrates that a digital twin may be used to improve livestock feed efficiency.

techniques for a ranch house in Fig. 4.

Results

https://github.com/AxDen-Dev/NRF52_3_Axis_Livestock_Motion_Tracker_example [22]

It's a Bluetooth Low Energy (BLE)-connected motion tracker. The NRF52811 SoC allows for Bluetooth Low Energy (BLE) connectivity. One can toggle the device on and off using a magnet and the included Si7201 Hall sensor. The KXTJ3 sensor is used to determine acceleration. The temperature is recorded and measured with the help of NTC. It uses a battery that lasts for five years. The movements of animals like cows and lambs may be categorized using sensors and artificial intelligence learning. The use of different sensors varies depending on the application. Generally speaking, sensors are used to measure physical quantities such as temperature, pressure, humidity, light, sound, and motion. The KXTJ3 sensor is used to measure acceleration, which is the rate of change of an object's velocity over time. NTC stands for Negative Temperature Coefficient, which is a type of thermistor used to measure temperatures. The movements of animals like cows and lambs may be classified using sensors and artificial intelligence learning, the focus of classification is not limited to the animal's movement. Factors such as environment, behavior, and physiology are taken into account. The sensor served as the cow's Ear Tag. We were able to master Edge AI, which divides a cow's daily routine into four distinct phases: meditating, eating, resting, and high activity. Edge AI motion classification helped us identify six diseases, including over-heating and illness.

Analysis of the temperature rate

By eliminating the transducer's injection site, the RFID device's temperature is low, associated with the veterinary thermometer's core temperature $t = 0.33$. By examining the variation created by the RFID's implant site, $F = 60.26$ was found, indicating that the transponder's temperature value is strongly impacted by the location. Retro-auricular correlation is $t = 0.627$. Eq. (13) determines the regression trajectory, although the multiple determination coefficients explain 94% of the RFID temperature variability. The statistics imply that the retro-auricular transponder temperature can accurately measure core temperature. The temperature will be analyzed using Eq. (13),

$$\begin{aligned} \text{Temperature humidity index} &= \text{temperature of air measured by a thermometer} \\ &- (0.55 - (0.55 \times \text{Relative humidity}/100)) \times (\text{temperature of air measured by a thermometer} - 58) \end{aligned} \quad (13)$$

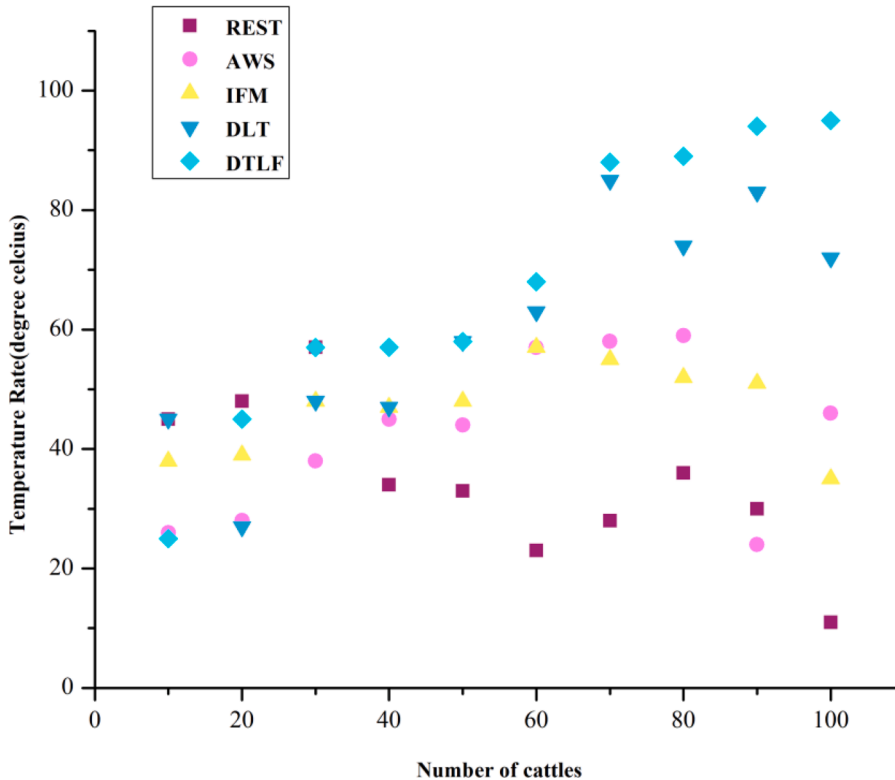


Fig. 5. compares the suggested study DTLF's temperature rate to that of previous research.

When testing an RFID sensor, it is necessary to take readings from both the sensor and the measuring instrument simultaneously. In this experiment, the cattle themselves serve as the thermometers. This evaluation makes use of a contactless thermometer as the measurement tool. All the examined cattle were healthy herding breeds with an average age of 1 year. In this experiment, the cattle are used to measure temperature changes. They are acting as thermometers. RFID, short for radio frequency identification, is a wireless system that uses tags and readers to identify individual objects. The RFID tag communicates with the reader, which is a device with one or more antennas that sends out radio waves and receives signals from the tag. This experiment aims to demonstrate that the value measured by the RFID sensor may serve as a reliable alternative to or complement the results of a traditional measurement. The infrared thermometer allows you to take many readings at once, making it ideal for checking an animal's temperature. In this experiment, an infrared thermometer takes readings by sending infrared rays into the cattle's body. At the same time, an RFID sensor is brought close to the cattle's body and reads its temperature after receiving the SMS command "Temperature." Additionally, the LCD infrared thermometer and mobile phones will show the values of the temperature monitoring data through SMS. The average discrepancy between the RFID sensor and the Infrared Thermometer is 0.8 °Celsius, which equates to an error value of 0.25% for the latter. There is a 0.9 C discrepancy between the RFID temperature reading and the infrared thermometer's reading. As a result, the RFID sensor provides 99% confidence when used to gage cattle core temperature. Fig. 5 compares the proposed DTLF to previous studies on precision livestock farming, such as the DLT, IFM, AWS, and REST architecture.

Analysis of relative humidity rate

$$\text{Relative humidity} = \frac{\text{Actual vapor pressure}}{\text{Standard vapor pressure}} \times 100 \quad (14)$$

The AM2302 is similar to the DHT22 but requires wires instead of batteries. As a humidity and temperature sensor, DHT22 excels in precision. This humidity monitor keeps tabs on the relative humidity levels in a room. The humidity is detected with the help of a capacitive sensor in Eq. (14). A non-contact thermistor (NTC) provides an accurate temperature reading. Despite its delicate appearance, this sensor can withstand extreme environments. You may get it as a sensor or a Module.

Regarding weather monitoring, the DHT22 sensor is superior to its predecessor, the DHT11. A sensor and a module, DHT22, may be used to measure temperature and humidity. The DHT22 sensor and module have similar operational principles. The interior workings are where things diverge. Both a filtering capacitor and pull-up resistors are included in the module. But with sensors, these connections are made outside. An 8-bit microcontroller is linked to both the DHT22 sensor and module. The sensor is available in a 4-pin package, whereas the DHT22 module only has a 3-pin one. The module outperforms the sensor in measurement range and accuracy

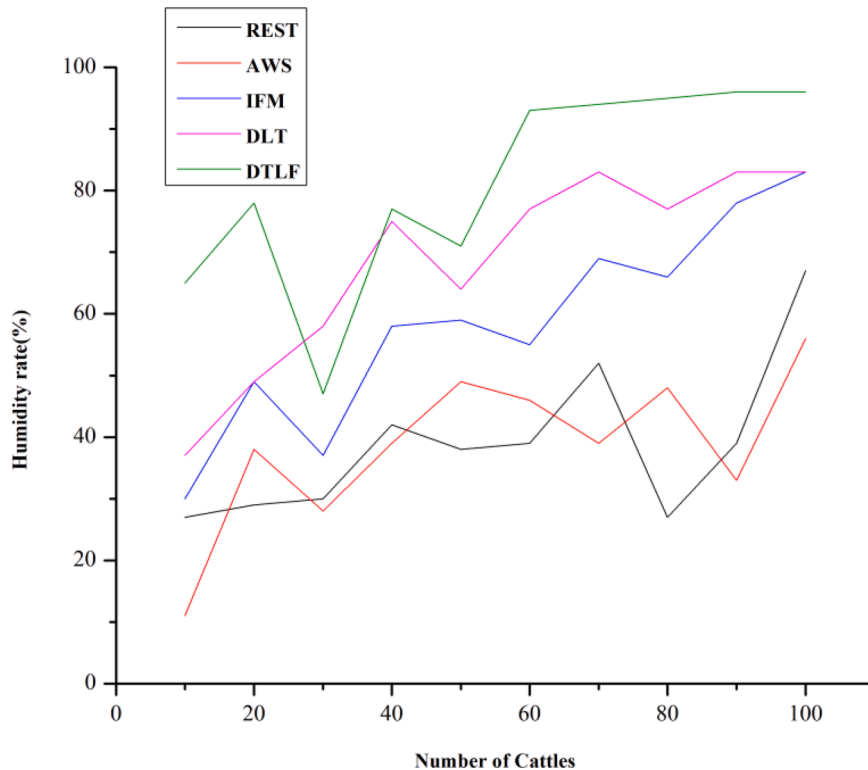


Fig. 6. compares the suggested study DTLF's humidity rate to that of previous research.

and costs more.

The Temperature Humidity Index (THI) is a method for evaluating the potential for heat stress in cattle and other animals by considering the combined effects of ambient temperature and relative humidity. Heat stress is triggered in cows when the THI is more than 82, which might harm the number of calves born to each cow. The milk from cows is severely diminished at pH 81. Milk production drops drastically, cows exhibit indications of extreme stress and some may even die at 72. THI values vary between cow species. Fig. 6 compares the proposed DTLF to previous studies on precision livestock farming, such as the DLT, IFM, AWS, and REST architecture.

Cattle have a normal body temperature of 36.5–39.5 °C. If a cow's body temperature is below normal, its milk supply drops 3–4 liters per day, and it may get illnesses including milk fever, indigestion, and poisoning. If it's above normal, it might contract anthrax, foot-and-mouth disease, and others. Room humidity is 57%–64%. Milk yield relies on humidity above usual. Daily milk consumption was reduced by four to six liters.

Heartbeat rate measurement

Adult cattle typically have a heart rate of 48 to 84 beats per minute. Photo Plethysmo Graphy (PPG) is a non-invasive way of detecting plasma capacity variation in tissues exploiting a sensor, and this approach uses the TCRT1000 reflecting optical sensor to accomplish so. This technique will be used to calculate the intestines since the increase in blood volume occurs in time with each pulse. PPG may be broken down into two primary types: transmission and reflection. Light is transmitted into the tissue and measured by a detector on the other side using a transmittance PPG.

For this reason, the transmission PPG can only be used on small areas, such as the tip of a finger or an ear lobe, where the sun may penetrate deeply enough to provide a useful effect. But in terms of PPG reflectivity, the light burden is rather small. The power source and the light sensor are located on the same side of the component. The detector reads the reflected light from inside the tissue. The lightweight does not need to be forced to enter the body. Hence the reflectivity PPG may be applied to almost any structural component. There may be a correlation between the observed light's pulsating intensity and the underlying pulsating blood flow induced by the outs' beating. Fig. 7 compares the proposed DTLF to previous studies on precision livestock farming, such as the DLT, IFM, AWS, and REST architecture. The following mean pulse rate data with pauses in the measurement period were obtained using such equipment on calves of various ages and sexes, maintained in a tethered rearing method. Cows between 560 and 610 kgs in body mass have a resting metabolic rate of 79 min per kilogram of body mass, cows between 610 and 670 kgs in body mass have a resting metabolic rate of 80 min per kilogram of body mass, and cows over 670 kgs in body mass during rumination have a resting metabolic rate of 51 min per kilogram of body mass. A sensor kept track of the animals' heart rates every four seconds for six minutes. The obtained mean heart rates throughout testing are approximate and do not represent the issues with the readings. The number of beats

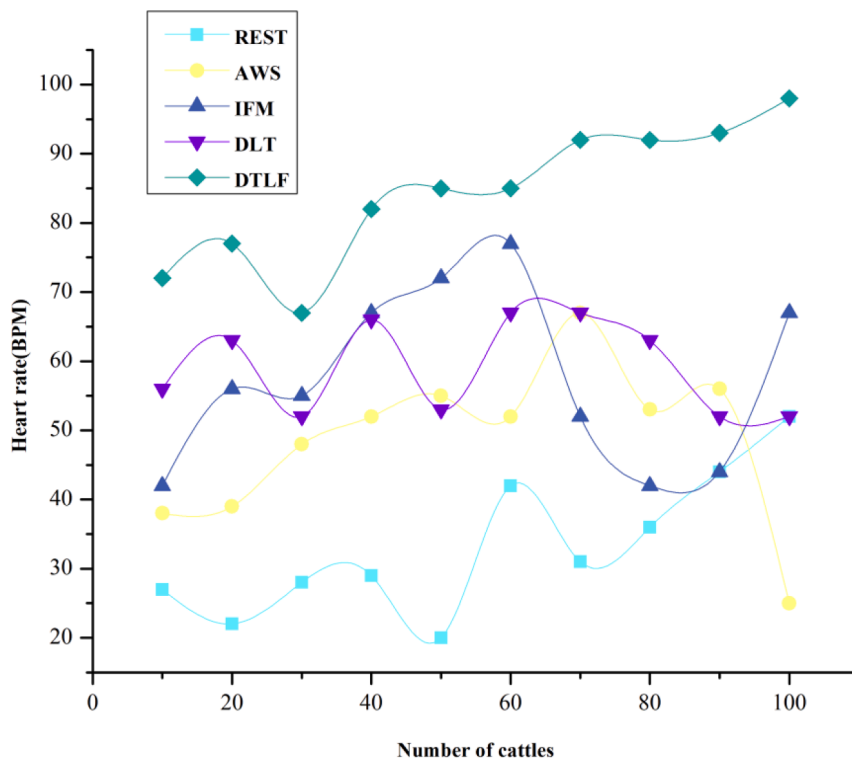


Fig. 7. compares the suggested study DTLF's heart rate to that of previous research.

per minute (BPM) is proportional to the frequency of these pulses in (15), as follows:

$$BPM = 60 * frequency \quad (15)$$

Analysis of earlier lumpy skin disease prediction

We built the model assuming that the farmer had a solid understanding of cattle conditions. When applied to cattle, the Digital Twins in Livestock Farming (DTLF) framework provides a thorough comprehension of cattle health and nutrition. It takes a comprehensive view of cattle management by incorporating ideas from animal nutrition, veterinary medicine, and animal behavior. Instead of treating individual symptoms, this model prioritizes addressing the animal's overall health. The DTLF considers the interplay between the animal, its environment, and the management techniques employed to solve the most immediate issues. The 98 symptoms for which the produced prediction is reliable and consumers would use the symptoms as input are all recognized. Data preparation refers to the mining technique of transforming raw data from datasets or data encoding into a form the algorithm can successfully translate. The purpose of data encoding algorithms is to convert raw data into a format that can be read and understood by another algorithm, such as a machine learning algorithm. Algorithms for data encoding can be used to convert information into a form that is more conducive to machine processing and interpretation. Encoding algorithms can be used to improve the quality of data in some situations. To make the data more suitable for a machine learning algorithm, one could use a data encoding algorithm to, say, transform text into numerical values.

- The study at hand employs the following preparation techniques for its data:

Inconsistencies in the data are repaired when data is cleaned, for instance, by filling in missing values. The term "data cleaning" means rectifying data by eradicating mistakes, duplicates, and other errors.

- A large data set makes research more challenging, so it is important to reduce it. As a result, we eliminate some of the independent variables (symptoms) that may not have much of an impact on the dependent variable (outcome) (disease). This analysis selects 98 out of 128 symptoms strongly correlated with certain illnesses.

Output (diseases) is derived once the algorithm has been trained on the dataset using the given techniques. It can identify and forecast the most likely sickness based on the user's diagnoses, which are processed by the generated set of rules. Most of the time, outstanding outcomes may be obtained with little effort using the scalable ML approach known as random forest. According to the Decision Tree, overfitting is the main flaw of decision tree algorithms. It's almost as though the tree remembered the data. Random

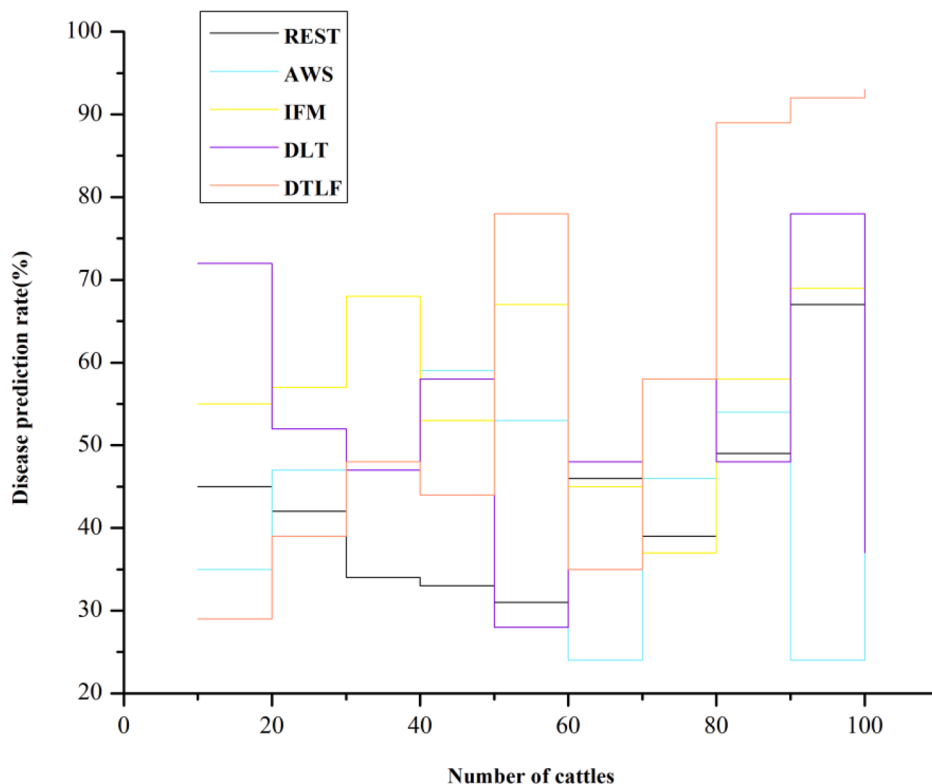


Fig. 8. compares the suggested study of DTLF's lumpy skin disease prediction rate to previous research.

Forest, a kind of ensemble learning, can avoid this problem. Some of the problems are over fitting occurs when a model is excessively complex, such as having too many branches and leaves. This causes the model to learn patterns from the training data that do not hold in the general population. Decision trees can be unstable because small changes in the training data can cause large changes in the structure of the tree. This can cause the model to over fit. It's relevant to using several algorithms or variants of the same algorithm. Random Forest is part of the Decision Tree family. More Random Forest decision trees mean more generality in (16).

$$\text{mean square error} = \frac{1}{M} \sum_{l=1}^M (gl - zl)^2 \quad (16)$$

Where M is the amount of data l , gl is the return value of the model, zl is the real value. Nave Byes is an algorithm that uses the Bayes theorem as its foundation (17).

$$P\left(\frac{t}{j}\right) = \frac{P(j/t)P(t)}{P(j)} \quad (17)$$

$P\left(\frac{t}{j}\right)$ is posterior likelihood, $P(j/t)$ is probability, $P(t)$ is probability of class, $P(j)$ and is probability of prediction. Reduced milk production may be attributed to certain illnesses in cattle. This study was conducted to forecast the onset of these illnesses by using supervised learning algorithms. Fig. 8 compares the proposed DTLF to previous studies on precision livestock farming, such as the DLT, IFM, AWS, and REST architecture.

The dataset may not behave as expected due to missing or noisy data. As a result, researchers need to use statistical methods to scrub the data. The models' effectiveness is measured by accuracy and confusion matrices. Data cleansing, pattern recognition, and the generation of predictive insights are all tasks amenable to statistical methods. Further applications for the statistical data include the creation of smart livestock environments that aid farmers in animal management, feed efficiency, and carbon footprint reduction. Statistical methods can also be used to analyze various aspects of animal health, such as disease diagnosis, herd health tracking, and welfare. As a result of this investigation, better decisions can be made regarding the animal's care and health, and a more complete picture of the animal's condition can be decorated.

This study demonstrates that cattle illness forecasting and prediction using data mining techniques is feasible. By using the model's output, veterinarians may foresee the onset of illness with greater precision.

Criteria for performance concept

The most widely-used performance indicators under study are briefly outlined below. These metrics provide a standard by which

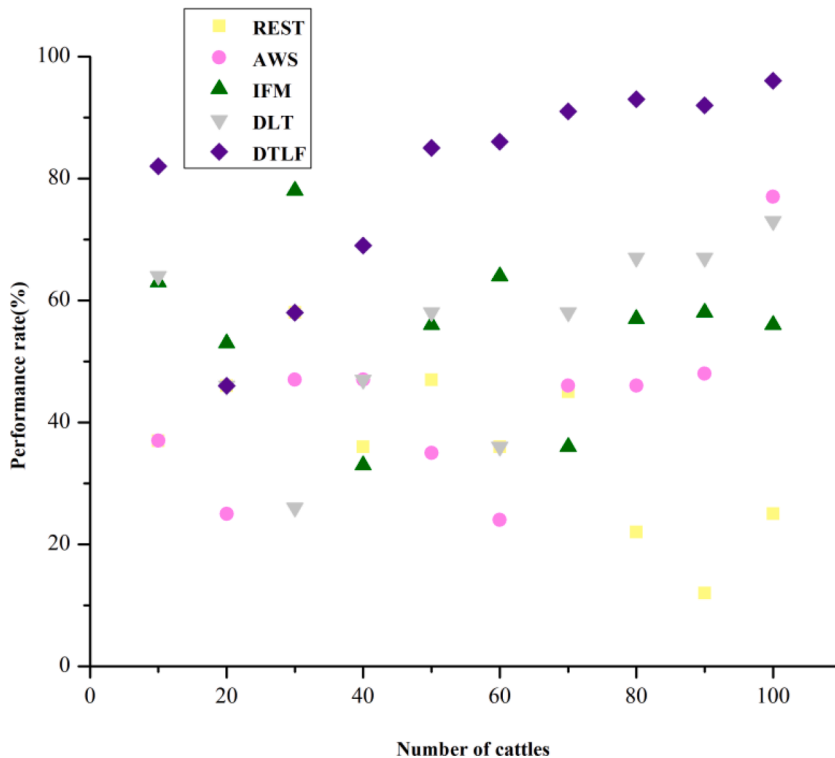


Fig. 9. compares the suggested study DTLF's performance rate to that of previous research.

ML algorithms can be judged. Choosing the right metrics is crucial because (a) individuals determine the algorithm's performance is evaluated, and (b) the metric itself can affect how the importance of various characteristics is weighted.

Fig. 9 compares the proposed DTLF to previous studies on precision livestock farming, such as the DLT, IFM, AWS, and REST architecture. The confusion matrix is one of the most natural measures of a model's accuracy. It is used to categorize issues when at least two classes may arise. Consider the simplest possible scenario: a target variable is set to "1" if a herd of cattle has been infected with a disease and "0" otherwise. This simpler version of the confusion matrix (Fig. 9) consists of a two-by-two table with columns labeled "Actual" and "Predicted" and rows that show the results of comparing the predictions with the actual class label. The formulas of the performance matrices are shown in Table 1. For the purpose of measuring the efficacy of a classification algorithm, a confusion matrix is commonly employed. It summarizes an algorithm's performance and shows the various kinds of mistakes it makes. Counts of correct predictions, incorrect predictions, and false negatives caused by the algorithm will all be displayed in the confusion matrix. In a smart livestock environment, for instance, a confusion matrix can be used to evaluate the efficacy of a classification algorithm for differentiating between healthy and sick livestock. This result may take on the following values in the preceding simplified example:

- The model correctly identifies this cattle as a diseased one (1), hence the diagnosis is correct (True Positive, TP);
- The model predicts that the cattle are healthy and returns a True Negative (TN) value of 0 if the illness is not present in the sample.
- In a false-positive (FP) scenario, the cattle do not suffer from the illness in question (value = 0), yet the model nonetheless identifies it as infected (value = 1);
- The cattle have a disease (1), yet the model indicates it is healthy (0).

Conclusion

This article provides a review of current research and development in cattle technologies that make use of the IoT. IoT and DTLF increased extensive PLF structure, technology, and equipment utilization. GPS, RFID, and computer vision can automatically track individual animals. As such, it covered topics such as cattle system design, platform, and topology that allow livestock health information and maintain and support the IoT infrastructure. This report also outlined recent developments in studies investigating the potential of IoT to check up on the wellness of cattle and the feed livestock consume. This article also provides a comprehensive overview of IoT livestock sensors and protocols changing cattle production. After thoroughly examining IoT-based livestock research, it is determined that more study is required to provide efficient cattle managing, tracking, monitoring, and prediction technologies.

In conclusion, this survey should be informative for academics, farmers, and legislators engaged in developing IoT-enabled livestock technology. As with other innovative concepts, the application of digital twin technology led to a 92% success rate in analyzing different farms for signs of illness. DTLF had a 92% success rate in measuring the heart rate, a 94% success rate in evaluating the temperature range, and a 94% success rate in evaluating the humidity range. Future possibilities of IoT-based DTLF for predicting the effects of climate change on cattle farming. Precision agriculture, in which AI-enhanced machines and robots manage crops and livestock without human intervention, may become part of farming's future scope. Technology has advanced to the point where farmers can use real-time data from machines and robots to make informed decisions about things like weather, soil quality, and animal health. Likewise, artificial intelligence-driven technology can be used to maximize harvests while decreasing expenses and boosting long-term viability. Artificial intelligence (AI) has the potential to monitor and manage livestock in geographically dispersed areas, develop and improve animal welfare systems, and increase the return from natural crops. The use of chemical pesticides can be decreased through the use of AI to keep an eye on and manage pests, and crop conditions can be tracked for maximum profit. AI can be used to automate processes and lessen the amount of manual labor needed on farms, freeing up time for farmers to devote to other endeavors.

Ethical approval

This article does not contain any studies with human participants or animals performed by any of the authors.

Author statement

Conception and design of study: **Shailendra Mishra**

Acquisition of data: **Sunil Kumar Sharma**

Analysis and/or interpretation of data: **Shailendra Mishra, Sunil Kumar Sharma**

Table 1
shows the formulas of performance metrics.

Performance Metrics	Formulas
Accuracy	$\frac{TP + TN}{FP + TP + TN + FN}$
Recall	$\frac{TP}{FP + TP}$
Precision	$\frac{TP}{FP + TP}$
F1 score	$(2 \times \text{RECALL} \times \text{PRECISION}) / (\text{PRECISION} + \text{RECALL})$

Declaration of Competing Interest

The authors declare that they have no conflict of interest.

Data availability

No data was used for the research described in the article.

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